

HOMEWORK ASSIGNMENT 4

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

Instructions:

- Kindly recall that homework assignments for this course are **not turned in**, and **not graded**.
 - However, problems from homework sets will form the basis of problems that appear on the weekly quizzes; as such, it is in your best interest to complete the homework sets on your own
 - Solutions to this assignment will be released in the morning of Friday September 18.
 - Kindly note that there will be **no quiz** on Monday September 21. However, the material this homework tests you on could absolutely show up on the **Midterm** on Wednesday September 23.
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1. **Harley's Return.** In this problem, we'll revisit the study from Lecture Activity 5 ("Can Dogs Understand Human Cues?") and reformulate it in the language of hypothesis testing. Recall the setup: Harley was presented with two cups, one of which the experimenter bowed towards. The research question was: can Harley understand human cues? The data was: Harley chose the "correct" cup (i.e. the cup the experimenter bowed toward) in 9 of the 10 total trials.

- (a) What proportion of Harley's trials resulted in him selecting the correct cup (what we are labeling as a "success")? Report the appropriate value, as well as its proper mathematical symbol.
- (b) What is the parameter of interest? Provide a verbal description in the context of the study, and also the proper mathematical symbol.
- (c) Write down the null and alternative hypotheses using the parameter you defined in part (b) above.

For Parts (c) onwards: Return to the One Proportion Applet, and conduct 10,000 repetitions of the chance model to obtain a simulated null distribution. Make sure that you have the option "Proportion of heads" selected (and not "Number of heads"). **Don't close out of your plot yet** - you'll be asked to modify various parts of it in the upcoming parts of this question!

- (d) What value did you use for the probability of success (π)?
- (e) What value did you use for the sample size (n)?
- (f) What does a dot on the simulated null distribution represent? Answer using the context of the study.
- (g) When trying to use the applet to approximate the p -value, should you use the $\leq$, $=$, or $\geq$ option in the applet? Explain.
- (h) Calculate your approximate p -value.
- (i) According to the guidelines posted in Lecture Activity 6 (and in the textbook), what is the strength of evidence?

- (j) Summarize the conclusion that you draw from this study based on your analysis. Make sure you explain the reasoning process behind your conclusion by referring to the (1) observed statistic, (2) approximate p -value (3) justify strength of evidence against the null model (very strong, strong, moderate, weak) based on the p -value, (4) reject/fail to reject the null hypothesis, and (5) overall conclusion in context. Make sure you refer to the **parameter** in your answer.

2. **Right-Handedness.** According to a somewhat old 2009 article (The Left Brain Knows What the Right Hand is Doing), 90% of people are right-handed. We'll take this to mean the authors claim that 90% of all people in the world are right-handed. Suppose we want to test this article's claim against the alternative that the true proportion of right-handed individuals in the world is less than 0.9. For our data, we'll consider the Welcome Survey I sent out earlier this semester to my three sections (S17, S18, and S19) of STAT 1110, in which 69 of the 84 respondents indicated they are right-handed.

- (a) Describe one observational unit in the context of this study.
- (b) Describe the population of interest, in the context of this study.
- (c) Describe the sampling frame, in the context of this study.
- (d) Define the parameter of interest. Be sure to use proper mathematical notation.
- (e) Write down the null and alternative hypotheses in terms of the parameter you defined in part (d) above.

Let's start off with a simulation-based approach. Return to the One Proportion Applet, and conduct 10,000 repetitions of the chance model to obtain a simulated null distribution. Make sure that you have the option "Proportion of heads" selected (and not "Number of heads"). **Don't close out of your plot yet** - you'll be asked to modify various parts of it in the upcoming parts of this question!

- (f) Calculate the approximate p -value.
- (g) Mark the correct options on the applet (it's up to you to remember what these are, from Lecture Activity 7) to calculate the mean and standard deviation of the 10,000 simulated $\hat{p}$ values.

In the remaining parts of this problem, we'll carry out a theory-based hypothesis test of the hypotheses posited in part (e) above.

- (h) Verify that the validity conditions are satisfied.
- (i) What are the center (mean) and standard deviation of the true/theoretical null distribution, and how can you tell?
- (j) How do your answers to parts (i) and (g) compare? Would they necessarily be *exactly* the same, or would we merely expect them to be *quite close* to each other?
- (k) Calculate the value of the standardized score.

- (l) Provide an interpretation of the standardized score you just calculated. **Hint:** think about how standardized scores represent the number of ____'s an observed statistic value is away from the ____.
- (m) According to the conventions established in Section 1.3 of the textbook, would we classify the observed value of the statistic as being in the tails of the null distribution? Explain briefly.
- (n) Based on your answer to the previous part, do you think there is sufficient evidence against the null to reject it in favor of the alternative? Why?
- (o) Navigate to the Theory-Based Inference Applet, and fill in the appropriate values. Click “calculate” and report the true/exact p -value. To show your work, include a screenshot of your entire screen showing the inputs and outputs.